PromoFlow — PRD (real version)

Oct 12, 2026 · Henry · brolab

Overview

The real version replaces every mock in the prototype with Gemini, Veo and server rendering, so a founder can turn a public YouTube Short into a Short for their own app in under 5 minutes.

- **Problem:** app founders know Shorts sell, but hiring a UGC creator costs about $180 a video, and copying a viral Short by eye misses the structure that made it work.
- **Product:** paste a Short that works → Gemini extracts its format (beats, timing, captions, CTA) → PromoFlow recommends the workflow and a shot list that format needs → it rebuilds the Short from the founder's own screen recording, screenshots and face clip.
- **User:** solo founders and small teams shipping apps or SaaS. Their demo is a screen recording, not a product photo.
- **Goal by Oct 18 (submission):** Gemini runs the format, script and demo cuts on the live URL.
- **Goal by Dec 4 (Demo Day):** accounts, server render, voiceover, Veo hooks and direct YouTube upload; real pilot founders publishing Shorts.
- **Non-goals:** long-form editing, AI avatars as the main product, TikTok and Instagram publishing (later).

Scope

One flow, end to end, with every step editable before the render.

1. **Sign in** with Google (also grants YouTube upload later).
2. **Paste a public Shorts link** → title, channel and thumbnail appear.
3. **Extract the format** → Gemini returns beats, durations, hook formula, caption style and CTA placement; cached per video ID.
4. **Get the recommended workflow** → PromoFlow builds only the blocks this format needs and wires them (no Proof block if the Short shows no results; a face-clip block if the creator speaks), plus a shot list per beat: what to record, what to say, which screen to show. The user accepts or edits it.
5. **Add the product by following the shot list:** context fields, 2–6 screenshots, one screen recording (≤ 3 min) with one action per step, the face-clip lines.
6. **Run** → script → hook clip → demo cuts → voiceover → proof → composed timeline. Each block shows its output as it finishes.
7. **Review** → edit any line, drag any cut, re-run one block without re-running the rest.
8. **Ship** → export a 1080 × 1920 MP4, or upload to YouTube as a Short with title, description and hashtags.

Out of scope for Dec 4: team workspaces, brand kits, multi-language voiceover, TikTok and Instagram publishing, analytics beyond YouTube views.

Functional requirements

The block graph and typed wires stay as in the prototype; what changes is what runs inside each block.

Block	Real behavior	Done when
Reference Short	Parse any Shorts/watch/youtu.be link; oEmbed title and channel	Private or invalid link shows a clear error in < 2 s
Format extractor	Gemini watches the Short from its URL; returns the format JSON; cached by video ID	Beat boundaries within ±0.5 s on ≥ 80% of the eval set
Workflow planner (new)	Maps each beat to the block that fills it and wires the graph; Gemini writes a shot list per beat from the product context	Every beat has a source block, no unused block; shot list shown before any upload
Product context	Name, description, audience, pain, result, features; optional “import from my website”	All fields feed the planner and script prompts
Screenshots, demo	Resumable upload to Cloud Storage; demo MP4/MOV/WebM ≤ 3 min, ≤ 500 MB	Upload survives a page reload
Script writer	Gemini writes one line per beat in the format's tone; uses only facts from context	Every step line ≤ 8 words; no claim absent from context
Hook clip	Upload a face clip recorded from the shot list, or generate with Veo 3.1 Fast (9:16, 4–8 s)	Veo clip ready in ≤ 3 min and labeled AI-generated
Demo cutter	Gemini finds the start and end of each step in the recording; user can drag to adjust	≥ 80% of cuts judged correct by a person
Voiceover (new)	Gemini TTS reads the script, timed to beats	Speech fits each beat within ±0.3 s
Proof collage	Grid from screenshots; optional real metrics typed by the user	Nothing invented on screen
Reel composer	Beat-ordered timeline, caption styles, Shorts safe zones, music bed	Browser preview matches the server MP4
Publish	Gemini writes title, description, #shorts tags; YouTube Data API upload via OAuth	Video appears on the channel as a Short

AI implementation

Six model calls make one Short. All return JSON checked against a schema, retry once, then fall back to the labeled mock so a run never breaks.

Step	Model	Input → output	Est. cost
Format	Gemini 3.8 Flash	YouTube URL → format JSON (beats, tone, captions, CTA)	~$0.01
Workflow + shot list	Gemini 3.8 Flash	Format + product context → block list and per-beat shot list (record, say, show)	< $0.01
Demo cuts	Gemini 3.8 Flash	Uploaded recording (Files API) + step list → `[{step, start, end}]`	~$0.01 per min
Script + metadata	Gemini 3.8 Flash	Format + product context → lines per beat, title, description, tags	< $0.01
Voiceover	Gemini TTS	Script lines → audio per beat	[verify price]
Hook clip (optional)	Veo 3.1 Fast	Prompt + optional first frame → 4 s 9:16 clip	$0.40

The block graph itself comes from a fixed beat → block mapping, so it is deterministic; Gemini only writes the shot list.

Guardrails

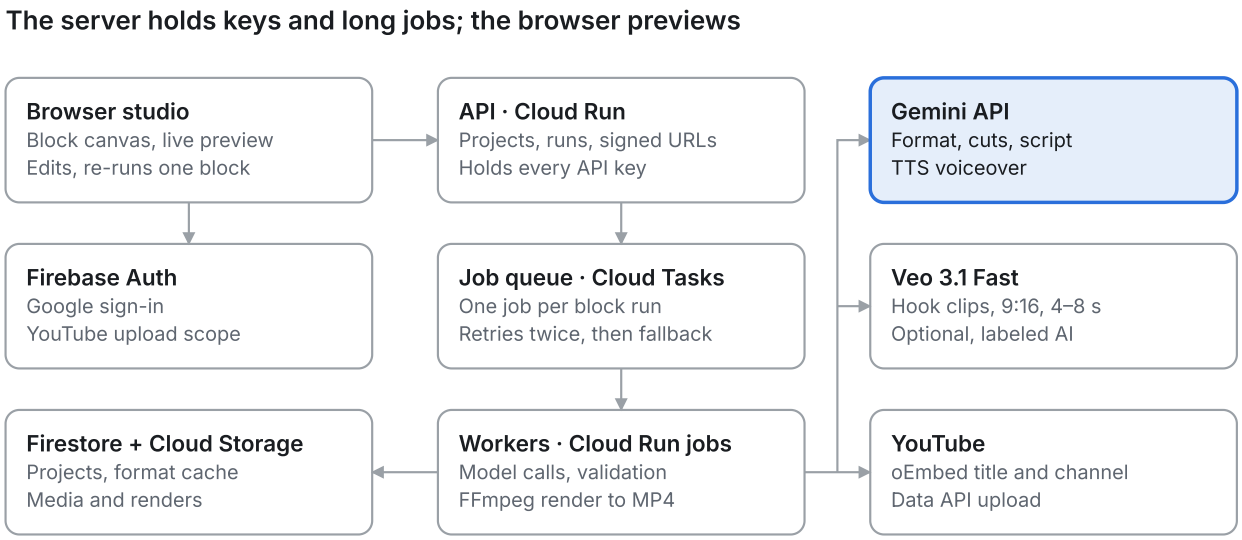
- Facts only from the product context; the prompt forbids invented metrics, users or quotes.
- Never reuse the reference's wording: reject a script line that shares 5+ consecutive words with the reference captions.
- Veo hooks are labeled AI-generated in the app and flagged as synthetic content on upload.

Evaluation

- Hand-annotate 30 app and SaaS Shorts (beats, times, kinds). Track beat-boundary error and beat-kind accuracy for every prompt change.
- Score demo cuts and scripts by two reviewers on 20 runs; ship a prompt only if it beats the last one.

Prices: Gemini 3.8 Flash at $0.75 / $3.75 per 1M input/output tokens until Dec 31, 2026, about 100 tokens per second of video (pricing, video docs).

Architecture and data



The studio stays in the browser for editing and preview; everything slow or secret runs on Cloud Run behind a job queue.

- **Stack:** keep today's vanilla-JS studio; Node API on Cloud Run; Firestore; Cloud Storage with signed-URL uploads; Firebase Auth.
- **Runs:** each block run is one Cloud Tasks job; its output lands in Firestore and the studio shows it as it arrives.
- **Render:** canvas preview in the browser; an FFmpeg worker renders the final 1080 × 1920 MP4 from the same timeline JSON.
- **Data:** Project (blocks, wires) · Format (videoId, JSON, model, date; cached) · Asset (owner, path, kind) · Run (block, status, output, cost).

Non-functional requirements

Area	Requirement
Cost	≤ $0.10 per Short without Veo, ≤ $0.60 with a Veo hook
Latency	Format ≤ 60 s; full run ≤ 3 min excluding Veo; preview starts ≤ 2 s after compose
Reliability	Every AI step has a labeled fallback; jobs retry twice; a failed block never blocks the rest
Limits	Gemini's YouTube input is in preview and free-tier capped at 8 h of video a day → paid tier + format cache
Privacy	Media is private to its owner, deleted on request and after 90 days of inactivity
Copyright	Only the structure of a reference is used; its video is never downloaded, stored or shown in output

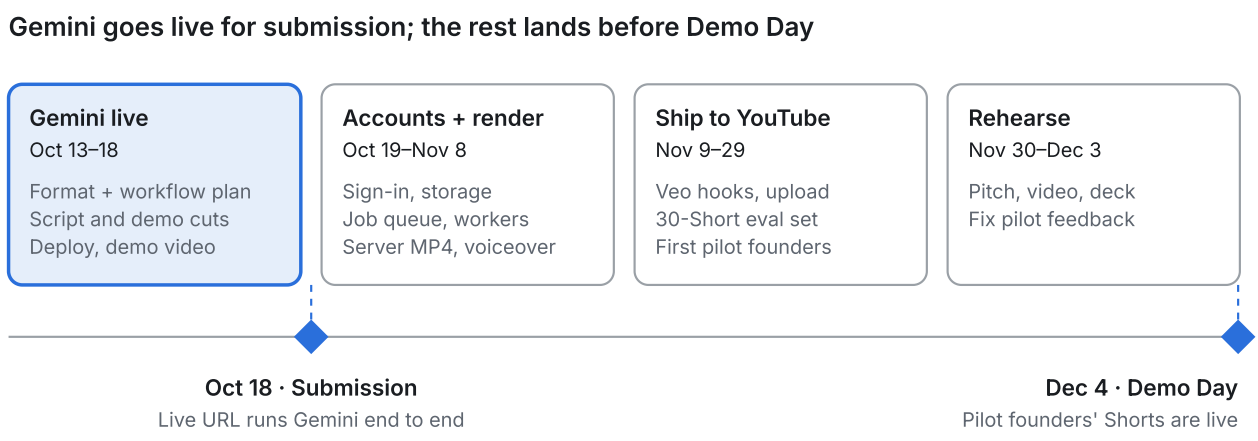
Area	Requirement
AI disclosure	Veo clips are labeled in the app and flagged as altered or synthetic content on YouTube upload
Browsers	Studio works in current Chrome, Edge and Safari; final MP4 renders on the server

Metrics

Targets for Dec 4 are proposals to agree on before the first sprint.

Metric	Target by Dec 4
Time from sign-in to first exported Short	≤ 5 min
New users who export a Short in their first session	≥ 40%
Format beat boundaries within ±0.5 s (eval set)	≥ 80%
Demo cuts judged correct	≥ 80%
Runs that finish without a fallback	≥ 95%
Model cost per Short, no Veo	≤ $0.10
Pilot founders who published a Short	[N]
Shorts published through PromoFlow, views tracked	≥ 30

Milestones, risks and open questions



Only the Gemini sprint gates the submission; accounts, render and YouTube upload land before Demo Day.

Risks

- Gemini's YouTube input is a preview feature → cache formats, keep the hand-annotated fallback, test the paid tier early.
- Cuts on dense desktop recordings miss the action → users can drag cuts; add auto-zoom on clicks if time allows.
- Veo is slow and costs $0.40 a hook → optional, runs in the background; the founder's own clip is the default.
- Scope vs a small team → Sprint 3 items are cut first; upload and voiceover outrank Veo.

Open questions

- □ Who owns frontend, backend and AI evaluation?
- □ Gemini API or Vertex AI, given the Google Cloud prize?
- □ Caption and voiceover languages at launch: English only, or Vietnamese too?
- □ Pricing to test with pilots: $19 / $49 a month, or per Short?